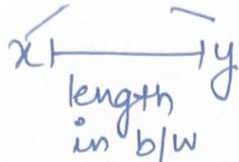


Metric Space

↓
R.-val (means distance function)
no.



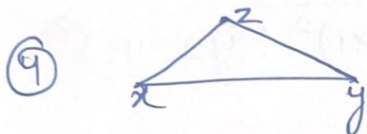
x & y is distance.

properties of distance.

① It is ^{not} negative.

② If pts are same $\leftrightarrow$ distance = 0

③ Symmetry $x \leftrightarrow y = y \leftrightarrow x$



$$\text{dist}(x, z) + \text{dist}(z, y) > \text{dist}(x, y)$$

Triangle Inequality follows.

$$d: X \times X \rightarrow \mathbb{R}$$

↑
dist functⁿ

let X be a non empty set

A metric (i.e; distance functⁿ) on X is a real functⁿ

$$d: X \times X \rightarrow \mathbb{R}$$

$$d(x, y) \rightarrow \mathbb{R}$$

(X, d) is metric space if it follows some properties
if d follows the above mentioned 4 properties
of metric / distance function)

$$① \quad d(x, y) \geq 0$$

$$② \quad d(x, y) = 0 \Rightarrow x = y$$

$$x = y \Rightarrow d(x, y) = 0$$

$$d(x, y) = 0 \Leftrightarrow x = y$$

$$③ \quad d(x, y) = d(y, x)$$

$$④ \quad d(x, z) + d(z, y) \geq d(x, y)$$

In $\mathbb{R}$ case

$$d = |x - y|$$

In $\mathbb{R}^2$ case

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Trivial / Discrete Metric Space

Let X be an arbitrary non-empty set $d: X \times X \rightarrow \mathbb{R}$
 be a function such that $d(x, y) = 0$ if $x = y$
 $d(x, y) = 1$ if $x \neq y$.

check $(X, d) \rightarrow$ metric space.?

① Non-negative $d(x, y) \geq 0$? $\{0, 1\}$ non-neg.

clearly $d(x, y) \geq 0$ ✓ Yes

② $d(x, y) = 0 \Leftrightarrow x = y$

$$\begin{aligned} &\rightarrow d(x, y) = 0 \Leftrightarrow x = y \\ &\rightarrow x = y \Rightarrow d(x, y) = 0 \end{aligned}$$

$$\sim q = \sim p$$

$$x \neq y \Rightarrow d(x, y) \neq 0$$

satisfied

(Already given
from defn
of d .)

③ Symmetry $d(x, y) = d(y, x)$

(1) $x = y$, $d(x, y) = 0$, $d(y, x) = 0 \Rightarrow d(x, y) = 0 = d(y, x)$

(2) $x \neq y$, $d(x, y) = 1$, $d(y, x) = 1 \Rightarrow d(x, y) = 1 = d(y, x)$

④ Triangle Inequality

$$d(x, y) \leq d(x, z) + d(z, y)$$

$d(x=y)$ $d(x \neq y)$

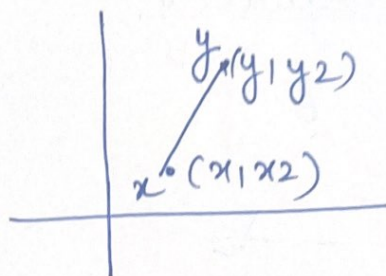
$\checkmark x=y=z (0=0+0)$ $x \neq y=z$ $1 \leq 1+0 \checkmark$

$\checkmark x=y \neq z$ $z=x \neq y$ $1 = 0+1 \checkmark$

$0 \leq 1+1$ $x \neq y \neq z$ $1 < 1+1 \checkmark$
 $0 < 2$ $1 < 2$

Euclidean metric space

$(\mathbb{R}^2, d)$
 denoted plane.



$x, y \in \mathbb{R}^2$

$x_1, x_2 \in \mathbb{R}$ $x = (x_1, x_2)$

$y_1, y_2 \in \mathbb{R}$ $y = (y_1, y_2)$

$(\mathbb{R}^2, d)$

$$d(x, y) = \left[(x_1 - y_1)^2 + (x_2 - y_2)^2 \right]$$

Euclidean distance.

$d : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R} / (0, \infty)$

① non negative holds.

② $d(x, y) = 0 \Leftrightarrow x = y$ holds.

$$[(x_1 - y_1)^2 + (x_2 - y_2)^2]^{1/2} = 0 \Leftrightarrow (x_1 - y_1)^2 = 0, (x_2 - y_2)^2 = 0$$

$$\Leftrightarrow x_1 - y_1 = 0, x_2 - y_2 = 0$$

$$\Leftrightarrow x_1 = y_1, x_2 = y_2$$

$$\Leftrightarrow (x_1, x_2) = (y_1, y_2)$$

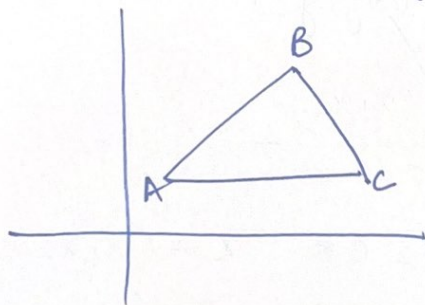
③ Symmetry: holds.

$$d(x, y) = d(y, x) \checkmark$$

$$d(x, y) = [(x_1 - y_1)^2 + (x_2 - y_2)^2]^{1/2}$$

$$[(y_1 - x_1)^2 + (y_2 - x_2)^2]^{1/2} = d(y, x)$$

④ Triangle Inequality



$$d(x, y) \leq d(x, z) + d(z, y)$$

$$[(x_1 - y_1)^2 + (x_2 - y_2)^2]^{1/2} \leq [(x_1 - z_1)^2 + (x_2 - z_2)^2]^{1/2} + [(z_1 - y_1)^2 + (z_2 - y_2)^2]^{1/2}$$

Use Minkowski's Inequality for $\mathbb{R}^3, \mathbb{R}^4$ and higher dimension.